

Zachary Hall

Stanford University
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Education

PhD in Philosophy, Stanford University (expected June 2026)

Dissertation Title: Undertheorized Aspects of Spin-Gravity Coupling

Advised by Professor Thomas Ryckman

Submitted for completion on Feb 26, 2026

Abstract: My dissertation brings out that the way the gravitational field and spin or intrinsic angular momentum dynamically relate to one another is significantly undertheorized by showing that there are *prima facie* many such couplings internally consistent and consistent with quantum field theory (QFT) and general relativity (GR). To show this, I first identify open empirical questions associated with such coupling. I then show that the Mathisson-Papapetrou-Dixon equations that model how macroscopic spinning objects couple to the gravitational field in GR do not bear on the issue despite the pervasive belief that they do by attention to an idealization inherent in them responsible for this belief. I also show that the “minimal coupling” procedure provides no *a priori* constraints on such theorizing. As an aside, I provide a new reconstruction of Bohr’s account of measurements of position and momentum (This is ultimately relevant for theorizing spin through theorizing measurements of spin, and Bohr’s account suggests ready generalization to this theoretical context.) Lastly, I consider whether and, if so, in what sense we should expect a successful theory of spin-gravity coupling to be background independent.

Patrick Suppes Fellow in the Philosophy of Science, September 2018-June 2023

Patrick Suppes Dissertation Fellow, September 2023-June 2024

B.A. in Astronomy & Physics and in Philosophy, Boston University, 2017. *Summa Cum Laude*

GPA: 3.99/4.00

College Prize for Excellence in Philosophy

College Prize for Excellence in Physics

College Prize for Excellence in Astronomy

Augustus Howe Buck Scholar

Area of Specialization

Philosophy of Science, Kant, Wittgenstein, Metaphysics, Philosophy of Physics

Areas of Competence

History of Modern Philosophy, History of Early Analytic Philosophy

Languages

English (native speaker)

German (conversational with advanced oral comprehension and reading levels)

Publications

“Bohr on EPR, the Quantum Postulate, Determinism, and Contextuality” *Forthcoming In Foundations of Physics* 54. No. 3, 1-35. (2025)

Kant, Nonsense, and the Suprasensible Reality of Thinking: Re-Reading Wittgenstein’s “Picture Theory” (with Robert Engelman). *Forthcoming in Wittgenstein-Studien*.

Proceedings

“A New Reading of the Second Analogy: The Nomologically-Determined Object View.” *Forthcoming in Christoph Horn & Rainer Schäfer, Kant’s Project of Enlightenment: Proceedings of the 14th International Kant Congress*. Berlin: De Gruyter.

Works In Preparation

Kant on Material Unity, Systematicity, and the Progressive Approximation of Nomic Necessity

A Development in the Dynamical Approach to GR and Emergent Spacetime Physics

Talks

Freedom and the Supersensible Reality of Thinking in Wittgenstein’s *Tractatus*.

Realismus und Idealismus heute III Kolloquium, Universität Bonn, 04. Mai, 2023

Research Colloquium of Markus Gabriel, University of Bonn, May 22, 2023

Forms of Thought: From Classical German to Early Analytic Philosophy, University of Stuttgart, July 1, 2024

Metaphysics and Freedom Conference presented by Stanford University’s German Philosophy Group, March 14, 2025

An Immanent Critique of Many-Worlds Interpretations of Quantum Mechanics, or: On the Reality of Quantum Jumps. Bonn History and Philosophy of Physics Research Seminar, May 16, 2023

Comments on “Grammar and Method in Wittgenstein’s Philosophical Investigations,” Pacific APA in Portland, OR, March 21, 2024

Non-Living Natural Kinds and the Material Unity of Nature in the Critical Philosophy, Kant and Hegel on Objectivity and Nature Workshop, Stanford University, November 16, 2024

Teaching Experience

Learning Assistant for CAS PY 105: *Physics 1* at Boston University, Fall Semester 2014

Learning Assistant for CAS PY 106: *Physics 2* at Boston University, Spring Semester 2015

Tutor for CAS PY 106: *Physics 2* at Boston University, Spring Semester 2015

Mathematics Teacher at St. Johnsbury Academy, 2017-2018

Teaching Assistant for PHIL 165: *Quantum Mechanics*, Winter Quarter 2020 at Stanford University

Teaching Assistant for PHIL 165: *Space and Time*, Winter Quarter 2021 at Stanford University

Teaching Assistant for *PHIL 150: Mathematical Logic*, Fall Quarter 2022 at Stanford University

Teaching Assistant for *PHIL 165: Philosophical Issues in Quantum Mechanics*, Winter Quarter 2022 at Stanford University

Teaching Assistant for *PHIL 80: Mind, Matter, Meaning*, Spring Quarter 2022 at Stanford University

Teaching Assistant for *PHIL 165: Philosophical Issues in Quantum Mechanics*, Winter Quarter 2024 at Stanford University

Teaching Assistant for *PHIL 28S: Philosophical Issues in Artificial Intelligence*, Summer Session 2024 at Stanford University

Humanities, STEM, and Writing Tutor at SARC, Summer Session 2024 at Stanford University

Primary Instructor for *PHIL 60S: Introduction to the Philosophy of Science*, Summer Session 2025 at Stanford University

Teaching Assistant for *PHIL 102: Modern Philosophy from Descartes to Kant*, Fall Quarter 2025 at Stanford University

Teaching Assistant for *PHIL 128: Kant and German Idealism*, Winter Quarter 2025 at Stanford University

Professional Service

Co-President, Graduate Study Body, AY 2019-2020

Faculty-Student Representative, AY 2020-2021

Professional Development Officer, AY 2022-2023

Co-organizer of Kant Retreat for Stanford Graduate Students and Postdoctoral Scholars, Summer 2022

Co-organizer of Kant Retreat for Stanford Graduate Students and Postdoctoral Scholars, Winter 2024

Co-organizer of Kant and Hegel on Objectivity and Nature Workshop, Stanford University, November 2024

Co-organizer of Conference “Metaphysics and Freedom” by Stanford’s German Philosophy Group, Spring 2025

Organizer of Stanford Philosophy Department’s Weekly Graduate Student Workshop, Spring 2025

Graduate Coursework

Philosophy of Science

Independent Study on the Gauge Principle in Quantum Physics – Winter 2019, Thomas Ryckman

Seminar on the Philosophy of Physics: On the Possibility of a Scientific Metaphysics – Spring 2019, Thomas Ryckman

Aristotle’s Physics Book One – Fall 2019, Alan Code

Scientific Philosophy: from Kant to Kuhn and Beyond – Spring 2020, Michael Friedman

Mental Action and its Pathologies – Spring 2020, Antonia Peacocke

Quantum Field Theory I (audit) – Fall 2021, Renata Kallosh

The Aim and Structure of Cosmological Theory – Winter 2022, James Weatherall

Philosophy of Mathematics and Logic

Mathematical Logic – Fall 2018, Ray Briggs

Seminar on Philosophy of Logic and Mathematics (Conventionalism) – Fall 2018, Jared Warren

Applied Group Theory – Fall 2018, Christine Taylor

Metalogic – Winter 2022, Thomas Icard

History of Philosophy

History of Ethics: Central Questions in Modern Ethical Theory – Winter 2019, Terence Irwin

Political Philosophy: the Social Contract Tradition – Winter 2019, David Hills

Kant's Third Critique – Spring 2019, David Hills

Aristotle's Physics Book One – Autumn 2019, Alan Code

Wittgenstein – Autumn 2019, David Hills

Kant's First Critique – Winter 2020, Michael Friedman

Scientific Philosophy: from Kant to Kuhn and Beyond – Spring 2020, Michael Friedman

Plato's Later Metaphysics and Epistemology – Winter 2021, Alan Code

Kant's Foundations of Morality: the Second Critique – Winter 2021, Michael Friedman

Wittgenstein (audit) – Winter 2022, David Hills

Value Theory

History of Ethics: Central Questions in Modern Ethical Theory – Winter 2019, Terence Irwin

Political Philosophy: the Social Contract Tradition – Winter 2019, David Hills

Kant's Third Critique – Spring 2019, David Hills

Truth – Winter 2020, David Hills

Metaphor – Spring 2020, David Hills

Kant's Foundations of Morality: the Second Critique – Winter 2021, Michael Friedman